

CONTACT INFORMATION Chicago, IL, USA, 60617
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RESEARCH INTERESTS

- **Planetary Habitability:** Earth-like exoplanet atmospheres and biosignatures
- **Exoplanet Spectroscopy:** Space-based transmission/emission spectroscopy of small exoplanets
- **Solar System Science:** Atmosphere and hydrological cycle systems of Titan

EDUCATION

University of Chicago, Chicago, IL

- PhD Student in Geophysical Sciences, Advisor: Prof. Edwin Kite, GPA: **4.0/4.0** 2023–
- M.S. in the [Physical Sciences Division](#) (GeoSci/Astro), GPA: **4.0/4.0** 2022
- Master’s Thesis: *The Role of Tectonic Luck and Weathering in Long-Term Climate Stability*

UCLA, Los Angeles, CA

2022–2023

- PhD Student in Planetary Science, Advisor: Prof. Jean-Luc Margot, GPA: **3.91/4.0**

UC Berkeley, Berkeley, CA

2015–2019

- B.A., [Astrophysics](#), GPA: **3.84/4.0**, *Cum Laude*, Dean’s List

EXTERNAL RESEARCH EXPERIENCE

- **NASA Jet Propulsion Laboratory**, Advisor: Dr. Keeyoon Sung Jul–Sept 2022
- **NASA Goddard Space Flight Center**, Advisor: Dr. Conor Nixon Jan–Jul 2021
- **NASA Marshall Space Flight Center**, Advisor: Danielle Moser Aug–Dec 2019
- **NASA Ames Research Center**, Advisor: Dr. Thomas Greene Jun–Aug 2019
- **UC Berkeley**, Advisor: Dr. Gaspard Duchêne Aug 2018–May 2019

FIRST-THIRD AUTHOR PUBLICATIONS

1. **Brandon Park Coy** et al. “Revealing the Shiny Nature of Lava Worlds: Small Planet Phase Curves with TESS”, 2026 (in review, *AJ*, arXiv:2609.26874)
2. **Brandon Park Coy**, Qiao Xue, Megan Weiner Mansfield et al., “Evidence for an Atmosphere on the Ultra-Short Period super-Earth HD 3167 b”, *ApJL*, 1005, L77, 2026
3. **Brandon Park Coy** et al. “Population-level Hypothesis Testing with Rocky Planet Emission Data: A Tentative Trend in the Brightness Temperatures of M-Earths”, *ApJ*, 987, 22, 2025.
4. **Brandon Park Coy**, Edwin Kite, and R.J. Graham. “The Role of Tectonic Luck in Long-term Habitability of Abiotic Earth-like Planets”, *PSJ*, 6(9), 218, 2025.
5. **Brandon Park Coy**, Conor Nixon, et al. “Spitzer IRS Observations of Titan as a Precursor to JWST MIRI Observations”. *PSJ*, Vol. 4 (6), p. 114, 2023.
6. Cole Smith, **Brandon Park Coy**, Megan Weiner Mansfield et al., “JWST MIRI Reveals a Potential Atmosphere on the Ultra-hot Rocky Planet TOI-431b”, accepted (*AJ*, arXiv:2609.03029), 2026
7. Sebastian Zieba, Laura Kreidberg, **Brandon Park Coy**, et al. “The dark and featureless surface of rocky exoplanet LHS 3844 b from JWST mid-infrared spectroscopy”, *Nat Astron*, 10, 1129, 2026.
8. Qiao Xue, Michael Zhang, **Brandon Park Coy**, et al. “The JWST Rocky Worlds DDT Program reveals GJ 3929b to likely be a bare rock”, *ApJL*, 995, L52, 2025.
9. Xuan Ji, Richard Chatterjee, **Brandon Park Coy** and Edwin Kite. “The Cosmic Shoreline Revisited: A Metric for Atmospheric Retention Informed by Hydrodynamic Escape”, *ApJ*, 992, 2, 2025.

	- Rafael Luque, Brandon Park Coy, Qiao Xue et al. “A dark, bare rock for TOI-1685 b from a JWST NIRSpec G395H phase curve”, <i>AJ</i>, 170, 49, 2025. - Conor Nixon, B Bézard*, T Cornet*, Brandon Park Coy*, I de Pater*, M Es-Sayeh*, H Hammel*, E Lellouch*, N Lombardo*, M López-Puertas*, J Lora*, P Rannou*, S Rodriguez*, N Teanby*, E Turtle*, et al., “Titan’s Atmosphere in Late Northern Summer from JWST and Keck”. (*co-second author) <i>Nat Astron</i>, 9, 969, 2025.
CONTRIBUTED PUBLICATIONS	- Courtney Dressing...Brandon Park Coy et al. “Early Exploration of the Scientific Discovery Space for the Habitable Worlds Observatory” (submitted, arXiv: 2608.11294) - Michael Zhang...Brandon Park Coy et al. “A carbon-rich atmosphere on a windy pulsar planet”, <i>The Astrophysical Journal Letters</i>, 995, L64, 2025 T. Bertin...Keeyoon Sung, Brandon Park Coy et al., “The HITRAN2024 methane update”, <i>JQSRT</i>, 349, 109736, 2025. - Megan Weiner Mansfield...Brandon Park Coy et al. “No Thick Atmosphere on the Terrestrial Exoplanet Gl 486b”, <i>ApJL</i>, 975, L22, 2024 - Qiao Xue...Brandon Park Coy et al. “JWST Thermal Emission of the Terrestrial Exoplanet GJ 1132b”, <i>ApJL</i>, 973, L8, 2024 - Jean-Luc Margot...Brandon Park Coy et al. “A Search for Technosignatures Around 11,680 Stars with the Green Bank Telescope at 1.15–1.73 GHz”. <i>AJ</i>, Vol. 166 (5), p. 206, 2023. - Gaspard Duchêne, J Oon, RJ De Rosa, P Kantorski, Brandon Park Coy et al., “A low-mass companion desert among intermediate-mass visual binaries: The scaled-up counterpart to the brown dwarf desert”. <i>MNRAS</i>, Vol. 519 (1), p. 778, 2023. - Edward Molter...Gaspard Duchêne, Brandon Park Coy et al. “Analysis of Neptune’s 2017 Bright Equatorial Storm”. <i>Icarus</i> Vol. 321 p. 324, 2019.
AWARDS/GRANTS	JWST Cycle 5 GO 10047 and 11694 Co-I, NASA/STScI (\$20,000) 2026 JWST Cycle 4 GO 7407, 7875, and 8005 Co-I, NASA/STScI 2025 Las Cumbres Observatory, 2025A Observing Proposal PI, <i>Starspot Characterization of HD 207496</i> 2024 Illinois Space Grant Fellowship, NASA/Illinois Space Grant Consortium (\$10,000) 2024 JWST Cycle 3 GO 6284 PI, 5263 Co-I, NASA/STScI (\$150,000) 2024 McCormick Fellowship, University of Chicago (\$4,500/year) 2023, 2024 Conference Travel Grant, STScI (\$2,500) 2023 Incoming Student Fellowship, UCLA (\$17,000) 2022
TALKS	Population-level Interpretation of Rocky Planet Surfaces and Atmospheres Seminar Max Planck Institute for Astronomy ExoCoffee Series 2026 Seminar Waterloo Centre for Astrophysics 2026 Seminar Carnegie EPL Astronomy Series 2026 Seminar STScI Exoplanets, Star, & Planet Formation Series 2026 Seminar University of Maryland Planetary Science Series 2026 Characterizing 10 Ultra-Hot Lava Worlds’ Thermal Emission with JWST Talk Other Worlds Laboratory Summer School (UCSC) 2026 Talk Exoplanets 6 (Porto, Portugal) 2026 Talk Rocky Worlds 4 Conference (Groningen, Netherlands) 2026 Atmospheric Theoretical Expectations for Rocky Worlds DDT Discoveries Invited Talk AAS 245 Rocky Worlds DDT Special Session 2025 Population-level Hypothesis Testing with Rocky Planet Emission Data: A Tentative Trend in the Brightness Temperatures of M-Earths

	◦ Talk Atmospheric Escape and Replenishment Workshop (STScI) 2025 ◦ Talk Other Worlds Laboratory Summer School (UCSC) 2025 ◦ Talk Atmospheric Characterization of Rocky to Giant Exoplanets in Thermal Emission 2025
	• What Sets the C/O Ratio in Sub-Neptune Atmospheres? ◦ Invited Talk Density Matters Workshop 2024 ◦ Lightning Talk Great Lakes Exoplanet Area Meeting 2023
	• Spitzer IRS Observations of Titan as a Precursor to JWST MIRI Observations ◦ Talk NASA Outer Planets Assessment Group Meeting 2021, 2023 ◦ Talk NASA Astrobiology Institute Titan Meeting 2021 ◦ Talk International Symposium on Molecular Spectroscopy 2021
TEACHING AND MENTORING	• Undergraduate Research Advisor to Olivia Lucal 2026- ◦ <i>Constraining albedo, atmosphere dynamics, and tidal deformation of hot Jupiters with TESS</i> • English and Math Instructor Elite Educational Institute, Seoul, Korea Jan 2020–Dec 2020 • Undergraduate Student Instructor (Astro 9 & C10) UC Berkeley Jun–Dec 2018
SERVICE	• UChicago Exoplanet Journal Club Organizer 2025–2026 • Rocky Worlds 4 Breakout Session Chair (“Airless Planetary Surfaces: Moon, Mercury, and Beyond the Solar System”) 2026 • Volunteer, South Side Science Festival 2026- • Volunteer, Adler Planetarium 2011-2015, 2026- • Executive Officer, OASES (Oakland Asian Student Educational Services), UC Berkeley 2015-2019 • Reviewer for the Astrophysical Journal